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Post-dural puncture headache prevention and treatment with aminophylline or theophylline: a systematic review and meta-analysis

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ABSTRACT / RESUMEN ESTRUCTURADO

Background:

Post-dural puncture headache (PDPH) is a common and debilitating complication of spinal anesthesia and lumbar puncture. Methylxanthines (aminophylline and theophylline) have been proposed as non-invasive therapeutic options. This systematic review and meta-analysis evaluated the efficacy and safety of aminophylline/theophylline in preventing and treating PDPH.

Methods:

A comprehensive literature search was conducted in PubMed, Embase, Cochrane Central Register of Controlled Trials (CENTRAL), and Scopus up to 2023. Randomized controlled trials (RCTs) evaluating intravenous aminophylline or oral theophylline versus control/placebo for PDPH prevention or treatment were included. Pooled standardized mean differences (SMD) and risk ratios (RR) were calculated using random-effects models.

Results:

Fifteen RCTs (1,240 total patients) met the inclusion criteria. Aminophylline/theophylline administration was associated with a statistically significant reduction in headache severity (SMD = -1.67; 95% CI: -2.28 to -1.05; P < 0.001) at 24 and 48 hours post-procedure. The incidence of PDPH was significantly lower in the intervention group (RR = 0.42; 95% CI: 0.28 to 0.62). Need for Epidural Blood Patch (EBP) was reduced by 64%.

Conclusions:

Aminophylline and theophylline are effective, safe, and cost-effective agents for both the prevention and treatment of PDPH, providing rapid pain relief and significantly reducing the requirement for invasive procedures such as epidural blood patch.

1. INTRODUCTION & PATHOPHYSIOLOGY

Post-dural puncture headache (PDPH) occurs in up to 40% of patients following accidental or deliberate puncture of the dura mater during spinal anesthesia or diagnostic lumbar puncture. The underlying pathophysiology involves persistent loss of cerebrospinal fluid (CSF) through the dural defect exceeding CSF production rate in the choroid plexus.

This fluid loss leads to intracranial hypotension and downward traction on pain-sensitive meningeal structures and cerebral blood vessels when the patient assumes an upright posture. Simultaneously, cerebral vasodilation occurs as a compensatory mechanism to maintain intracranial volume (Monro-Kellie doctrine), causing severe postural throbbing headache.

While conservative measures (bed rest, hydration, analgesics) and Epidural Blood Patch (EBP) remain traditional mainstays, methylxanthines offer a direct mechanism to counteract vasodilation and stimulate CSF production.

2. MECHANISM OF ACTION OF METHYLXANTHINES

A) Adenosine Receptor Antagonism (A1 / A2):
Adenosine induces potent cerebral vasodilation. Aminophylline and theophylline competitively block adenosine receptors, triggering active compensatory cerebral vasoconstriction and rapid headache alleviation.

B) Phosphodiesterase (PDE) Inhibition:
Inhibition of PDE enzymes increases intracellular cyclic adenosine monophosphate (cAMP), modulating pain pathways and vascular tone.

C) Stimulation of Na+/K+ ATPase Pump:
Methylxanthines stimulate active transport in the choroid plexus epithelium, enhancing active CSF secretion and restoration of intracranial pressure.

3. META-ANALYSIS RESULTS & FOREST PLOT SUMMARY

Outcome Measure	No. RCTs	Sample Size	Effect Size (95% CI)	P-Value
Pain Severity at 24h (VAS)	12 RCTs	980 pts	**SMD = -1.67 (-2.28, -1.05)**	**< 0.001**
Pain Severity at 48h (VAS)	10 RCTs	820 pts	**SMD = -1.42 (-1.95, -0.89)**	**< 0.001**
Incidence of PDPH (Prevention)	7 RCTs	640 pts	**RR = 0.42 (0.28, 0.62)**	**< 0.001**
Requirement for EBP	6 RCTs	510 pts	**RR = 0.36 (0.19, 0.68)**	**0.002**
Adverse Effects (Nausea/Tremor)	11 RCTs	920 pts	**RR = 1.12 (0.78, 1.61)**	**0.54 (NS)**

4. RECOMMENDED DOSING PROTOCOLS

INTRAVENOUS AMINOPHYLLINE PROTOCOL:

ï¿½ Dose: 240 mg to 480 mg IV diluted in 100-250 ml Normal Saline 0.9%.
ï¿½ Rate: Slow intravenous infusion over 30 to 45 minutes.
ï¿½ Frequency: May repeat every 12 hours if pain persists (maximum 2 doses in 24h).
ï¿½ Precautions: Monitor heart rate (avoid in severe tachycardia or arrhythmia).

ORAL THEOPHYLLINE PROTOCOL:

ï¿½ Dose: 250 mg orally every 8 hours for 2 to 3 days.

5. DISCUSSION & CLINICAL IMPLICATIONS

The findings of this meta-analysis confirm that intravenous aminophylline and oral theophylline are significantly superior to placebo and standard conservative therapy for PDPH.

Key takeaways for clinical practice:
1. Rapid Onset: Pain reduction is typically observed within 30 to 60 minutes after beginning the aminophylline infusion.
2. Reduced Invasive Interventions: Epidural Blood Patch (EBP) is an effective but invasive procedure with potential risks (infection, dural re-punciï¿½n, back pain). Methylxanthines provide a highly effective non-invasive bridging or primary therapy that prevents EBP requirement in over 60% of cases.
3. Safety Profile: Mild side effects (transient flushing, mild tachycardia, nausea) occur with similar frequency between groups and require no intervention.
4. Cost and Availability: Aminophylline is widely accessible in emergency rooms and surgical suites globally at minimal financial cost.

In conclusion, aminophylline should be considered a standard first-line pharmacological treatment for PDPH prior to performing epidural blood patch.

6. KEY REFERENCES

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